RESEARCH ASSESSMENT / TECHNICAL AND SCIENTIFIC REVIEW

Alash Science Bench

Strengths, limitations, validation strategy, and research roadmap

29 October 2026 • Repository: almubdimutaikhan/alash-science-bench

Overall verdict
Alash Science Bench is a promising, source-grounded research prototype, *not yet a scientifically validated benchmark*. Its most convincing feature is item-level provenance; its most consequential limitation is that the items have not yet been shown to measure the intended capability—knowledge of historical Kazakh scientific terminology and scholarly content—rather than general scientific knowledge, language inference, or multiple-choice cues.
Immediate recommendation: validate and experimentally test the existing 160 factual targets before scaling the dataset.

160	320	8	12
factual question–answer targets	prompts in two formats	scientific / educational areas	source volumes

Executive assessment

Dimension	Assessment
Research motivation	Strong: historically distinctive and underrepresented material
Provenance and traceability	Strong: explicit source and passage metadata
Engineering and documentation	Strong for a pilot: validators, review artifacts, exporter
Question diversity	Weak: dominated by terminology recall
Human gold-standard validation	Not established
Construct validity	Major unresolved question
Model evaluation	Proposed; no completed comparative results documented
Publication readiness	Premature as a validated benchmark paper

These are qualitative judgments based on the repository documentation, not numeric benchmark scores.

Scope. This is a research and methods review based on repository documentation, reported audits, and representative items. The evaluation scripts were not executed for this review, and the original private OCR materials and historical scans were not independently checked. Source and related-work links are given at the end.

1 Confirmed strengths

1.1 Source-level provenance and auditability

The dataset stores author and work identifiers, page references, OCR excerpts, character offsets, source hashes, and AI-review metadata. The documented validation pipeline checks whether quoted excerpts match the source OCR and whether updates invalidate stale reviews. This makes errors traceable and facilitates targeted correction. It is a substantive advantage over an opaque set of model-generated questions.

Qualification. Source checks cannot be independently reproduced from the public repository alone where the original OCR files or scans are unavailable. Traceability within the authors' workflow should not be confused with full public reproducibility.

1.2 Distinctive historical content

The target is not generic school science in Kazakh but historically attested scientific terminology and expressions in Alash-era scholarly works. This may expose failures that broad modern-language benchmarks miss, especially when a model understands a scientific concept but cannot identify its historical Kazakh expression.

1.3 Useful paired evaluation settings

The design contemplates two response formats (short answer and multiple choice) and two evidence conditions (closed book and supporting passage supplied). Together these could separate four behaviors:

	No supplied passage	**Passage supplied**
Short answer	Unassisted recall	Evidence-grounded generation
Multiple choice	Recognition / selection	Evidence-grounded selection

These conditions should not be collapsed into a single score: a model that selects a term from options is not necessarily able to produce it, and a model that answers after reading the passage need not possess the knowledge beforehand.

1.4 Documented limitations rather than inflated claims

The repository acknowledges that questions were AI-generated, AI-reviewed, and not yet human-validated. Its internal audit flags item-level concerns, including answer leakage and cross-question dependencies. This makes the project a credible starting point for systematic refinement.

2 Critical weaknesses and threats to validity

2.1 Construct validity: what does a correct answer actually show?

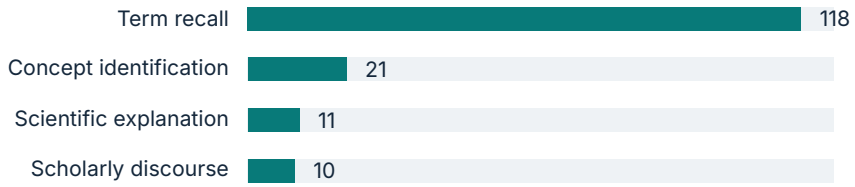
This is the central scientific issue. A model may answer a historical terminology item correctly because it has encountered the source, because it understands ordinary modern science, because the historical term is linguistically transparent, or because the options reveal the correct answer. One observed score cannot identify which mechanism produced success.

> **Main inference to avoid.** High accuracy on historical-source questions is not, by itself, evidence that a model has learned the Alash-era corpus or understands an author's specific scholarly position. Similarly, low accuracy may reflect orthographic unfamiliarity or question design instead of absent conceptual understanding.

Remedy: matched control conditions. For each focal concept, construct a modern-science question, a historical-term recognition item, a historical-term generation item, and (where justified by the source) a question about the author's explanation or use of that concept. Attribution controls can probe whether the model distinguishes sources rather than merely guessing a term.

2.2 Task imbalance and insufficient depth

The first-pass audit reports the following distribution across 160 targets:



118/160 (73.75%) targets are terminology recall. The 320 prompts are alternative renderings, not 320 independent factual questions.

A benchmark that predominantly measures term recall should not be presented as broad scientific reasoning or comprehensive Alash scholarship. A more coherent structure is to report two distinct tracks: **historical terminology** (recognition, recall, lexical variants) and **scholarly understanding** (explanation, interpretation, attribution, comparisons).

2.3 Author, discipline, and source are confounded

The project balances eight disciplines numerically at 20 targets each, but certain disciplines depend heavily on one author or work. As documented, physics and geometry rely on Eldes Omaruly while linguistics draws on Baitursynuly. A difference between subject accuracies can therefore reflect author familiarity, source visibility, language style, or task difficulty. Until source coverage is expanded, report outcomes as *source-specific subsets*, not representative estimates of discipline-wide competence.

2.4 Multiple-choice artifacts and question dependence

Some alternatives can be eliminated without historical knowledge; other items contain linguistic overlap between stems and correct answers. The audit further notes 114 distractor positions that contain the correct answer to some other item, creating opportunities for cross-question leakage when a set is presented jointly. Balanced answer letters do not resolve these concerns.

- Have reviewers annotate whether an answer is inferable without source-specific knowledge.
- Replace trivial distractors with plausible scientific or historical alternatives.
- Shuffle answer order and compare accuracy; consider an options-only diagnostic.
- Evaluate items independently and never expose the entire answer key to the same model run.

2.5 No human-verified gold standard

Automated excerpt matching and an independent model review are useful checks, but correlated interpretation errors remain possible. Historical orthography, OCR noise, terminology ambiguity, and translated works add further risk. Review should explicitly separate (i) mechanical traceability, (ii) linguistic and historical attribution, and (iii) scientific correctness and answer uniqueness.

Minimum workflow: double-review a 16-item calibration sample, refine instructions, then review the remaining set with explicit adjudication and a record of unresolved cases. Human reviewers should see the original page or a reliable facsimile, not only generated question text.

2.6 Small sample and non-independent observations

Twenty items per discipline imply a 5-percentage-point change from a single answer. Items also share authors, books, pages, and underlying concepts. Confidence intervals computed as if every prompt were independent would exaggerate precision. Model comparison should use paired predictions at the factual-target level and uncertainty methods sensitive to shared sources, while recognizing that only 12 source volumes may limit source-cluster inference.

2.7 Rights, release policy, and incomplete reproducibility

The public release contains OCR-derived passages while redistribution permissions are not fully established, according to the repository's rights note. A per-volume rights decision is therefore a release

prerequisite. The repository also lacks a finished, version-pinned multi-model evaluation with predictions, prompts, scoring rules, and results. These gaps should be addressed before publication claims about benchmark performance.

3 Experiments that would strengthen the scientific contribution

3.1 Experiment A: isolate the historical terminology gap

Question. Does a model understand a modern scientific concept yet fail to recognize or produce the historically attested Kazakh term?

> **Matched example (illustrative).**
> *Modern concept:* What is the role of a catalyst in a chemical reaction?
> *Historical term:* Which term is used for a catalyst in the cited historical Kazakh chemistry text?
> *Source interpretation:* What evidence in the passage supports that alignment?

Create concept-matched item families rather than unrelated pools of modern and historical questions. A within-concept gap helps distinguish missing historical terminology from missing scientific knowledge. Report the gap by model, item type, source, and answer direction; audit whether modern and historical prompts have comparable difficulty.

3.2 Experiment B: script and orthographic robustness

Question. How much does model performance depend on the writing system or representation of the same term?

Compare historically justified orthographic forms or verified transliterations using semantically identical prompts. Record whether each variant is an original spelling, a diplomatic transcription, a normalized spelling, or a modern translation. Do not interpret a spelling-induced performance difference as historical knowledge loss until the semantic equivalence and readability of the variants have been checked by knowledgeable reviewers.

3.3 Experiment C: historical evidence as an intervention

Question. Does supplying the supporting historical passage close the performance gap?

For each target, evaluate both closed-book and evidence-provided settings under a fixed prompt protocol. Compare the gain separately for terms, concepts, and explanations. A large gain indicates that the supplied information enables better answers; a small gain could reflect OCR quality, comprehension difficulty, ambiguity, or grading problems. Neither outcome, alone, identifies what a model memorized during training.

3.4 Recommended evaluation protocol

1. **Fix the dataset version.** Preserve source, item ID, prompt, answer rubric, and all variants under one release tag.
2. **Separate tracks and conditions.** Report short-answer and multiple-choice scores separately, with and without source passages.
3. **Use credible baselines.** Include the 25% uniform-choice baseline for four-way MCQ, several distinct multilingual/Kazakh-adapted models, and a modest human baseline if feasible.
4. **Avoid grading drift.** Write acceptable-answer rubrics before evaluation. Blind reviewers to model identity, and manually audit automated short-answer judgments.
5. **Report uncertainty and errors.** Use paired item-level comparisons, transparent interval estimates, disagreement analyses, and representative errors. Do not over-interpret small subject-level differences.
6. **Protect held-out evaluation.** Keep prompt variants for one factual target together. If prompts or models were optimized on the public pilot, treat that pilot as development data and evaluate on a newly curated source-disjoint set.

4 Prioritized implementation roadmap

Rank	Action	Deliverable / stopping condition
P0	Resolve OCR excerpt redistribution rights	Documented per-volume public-release policy
P0	Double-review 16 representative targets against original source pages	Calibration disagreements; revised annotation instructions
P1	Define the construct and task taxonomy	Distinct terminology and understanding tracks; scoring rubric
P1	Validate and repair all 160 factual targets	Reviewed pilot v0.2 with adjudications, rejected items, and metadata
P1	Add matched modern/historical controls	Pre-specified construct-validity experiment
P2	Implement version-pinned evaluation runner	Prompts, model versions, seeds, raw outputs, parsers, and metrics
P2	Evaluate baseline systems and inspect errors	Paired comparisons, uncertainty, and item-level analysis
P3	Collect additional authors and source volumes	Broader coverage and a genuinely held-out source-disjoint test set
P3	Add scholarly explanation and interpretation tasks	A broader second-track evaluation with human-reviewed answers

Recommended milestone. Complete a verified 160-target pilot and a small, interpretable baseline study before increasing volume. Dataset expansion is warranted only when the present items have demonstrated adequate clarity, discrimination, and source-based validity.

5 Novelty, publication potential, and project boundaries

5.1 Position relative to other Kazakh benchmarks

Existing work such as *KazMMLU*, *KazBench-KK*, and *KazCulture* addresses broader educational knowledge or cultural competence in Kazakh. Alash Science Bench's plausible niche is narrower: historically attested scientific expressions and source-specific scholarly terminology.

That niche is potentially useful but not sufficient by itself to establish a publishable methodological or empirical contribution. A stronger paper would test a proposition of the following form:

> **Testable research hypothesis.** Language models may exhibit a measurable gap between modern scientific concept knowledge and historically attested Kazakh scientific terminology; the magnitude of that gap may depend on answer format, orthography, source context, and model family.

This is a hypothesis, *not a finding*. The contribution would emerge from carefully controlled experiments, validated historical sources, and informative analysis, not simply from collecting additional AI-generated questions.

5.2 Relationship to a broader computational history of Kazakh

The benchmark measures *model knowledge of historical scholarly material*. It does not directly measure how the Kazakh language itself has evolved. Diachronic linguistic research would additionally require comparable corpora across periods, explicit normalization decisions, and analysis of word frequencies, syntax, register, orthography, and semantic change.

Alash Science Bench could remain a focused subproject and contribute a curated terminology lexicon, provenance database, and downstream evaluation track to a wider historical-Kazakh research

program. Its scope should not be expanded merely by renaming it a comprehensive language-change study.

6 Conclusion

The engineering and documentation are unusually well structured for a pilot of this size, especially the source-linked records and self-critical audit. The principal scientific gaps are independent expert validation, separation of target capability from easier shortcuts, and model experiments that produce interpretable comparisons.

Decision: prioritize validation and experimental design over scale. If the pilot establishes reliable historical terminology gaps, it can support a focused empirical contribution. If the tests expose easy guessing, ambiguous terminology, or source-dependent artifacts, that result should guide the next version rather than be hidden by collecting more items.

A Illustrative items from the chemistry subset

CHE01 – historical term recall (concept to term)

Қошке Кемеңгерұлы аударған «Кімійе» оқулығында реакцияны тездетіп, өзі өзгермей қалатын катализатор қандай атаумен берілген?

Gold answer documented in review: Тектестіргіш.

CHE04 – historical term interpretation (term to concept)

Қошке Кемеңгерұлы аударған «Кімійе» оқулығында «ішкі өзгеріс» қазіргі қандай құбылысқа сәйкес келеді?

Gold answer documented in review: Химиялық өзгеріс.

These reproduce representative examples cited in the repository assessment. They have not been independently verified against original scans in the preparation of this report.

B Source documents and reading

1. **Alash Science Bench (repository).** https://github.com/almubdimutaikhan/alash-science-bench
2. **First-pass review and audit.** https://github.com/almubdimutaikhan/alash-science-bench/blob/main/docs/first-pass-review.md
3. **Project next steps.** https://github.com/almubdimutaikhan/alash-science-bench/blob/main/docs/next-steps.md
4. **Rights and redistribution notes.** https://github.com/almubdimutaikhan/alash-science-bench/blob/main/RIGHTS.md
5. **Related benchmarks for positioning:** KazMMLU, KazBench-KK, and KazCulture. These are comparative reference points, not sources used to compute any of the project's reported statistics.

> **Evidence boundary.** Descriptive dataset counts, task composition, author/source confounds, and known audit issues are reported from the repository's own documentation. All recommendations, priority labels, and judgments about research validity are analytical interpretations. No original OCR corpus, historical page image, model evaluation log, or independent legal rights review was supplied for verification in this report.